

PS_Prod α Component Design Rationale — Why NonProd Headroom Drives Prod Placement

Document Type: Design Rationale / Technical Reference\

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Related Requirements: PLC-001, PLC-002, CAP-013, QUA-004\

Status: Design-of-Record

Executive Summary

The Production Placement Score (PS_Prod) uses **NonProd CRG headroom** in its α (availability fit) component, not Prod CRG headroom. This is a deliberate, forward-looking design choice that makes Prod placement a **multi-dimensional regional health check**, not just a “does this region have Prod capacity right now?” gate.

The Formula:

$$PS_Prod(r) \propto \text{component} = 0.30 \times \text{Clamp}(\overset{\uparrow}{\text{nonprod_crg.effective_free}} / \overset{\uparrow}{\text{prod_crg.quantity}})$$

NonProd FREE Prod REQUIREMENT

NOT:

✗ $0.30 \times \text{Clamp}(\text{prod_crg.effective_free} / \text{prod_crg.quantity})$ ← This would be self-referential

This document explains the five strategic reasons behind this design.

Five Strategic Reasons

1. Avoid Circular Self-Reference

If α used Prod headroom:

- You’d be evaluating “How much Prod capacity does this region have to accommodate this Prod workload?”
- This is **self-referential** — you’re measuring the thing you’re about to allocate against itself
- Creates a feedback loop where Prod placement depends on Prod capacity, which changes after Prod placement

With NonProd headroom:

- You're measuring an **independent signal** — NonProd capacity health is not directly affected by this Prod allocation
- Provides a forward-looking indicator of the region's overall capacity health

2. Future CVAL Co-Location Readiness (2-Region Model)

In a 2-region model (e.g., EU: West Europe + North Europe):

- **Prod** goes to Region A
- **CVAL + DR** co-locate in Region B (the other region)

The scoring logic anticipates this:

- If Prod goes to a region with good NonProd headroom, it signals that region can also support future NonProd (CVAL) growth in the same region if needed
- Even though CVAL typically goes to the OTHER region, having NonProd headroom where Prod lives provides overflow capacity and regional balance

Example — EU 2-Region Model

Scenario	Region A (Prod Candidate)	Region B (CVAL/DR Candidate)
Unbalanced	Prod Free = 8000 NonProd Free = 500 (tiny)	Prod Free = 2000 NonProd Free = 10000
Balanced	Prod Free = 5000 NonProd Free = 6000	Prod Free = 5000 NonProd Free = 6000

If α used Prod Free:

- Region A would score higher (8000 > 5000)
- But it's a **capacity trap** — Region A has almost no NonProd headroom; if any NonProd workload needs to overflow there, it fails

With NonProd Free:

- Balanced region scores higher (6000 vs 500)
- Prod placement favors regions with **holistic capacity health**, not just Prod-specific capacity

3. Regional Capacity Health Indicator

NonProd headroom is a **proxy for overall regional health**:

A region with ample NonProd headroom likely has:

- Good overall capacity management
- Room for future growth across all environments
- Lower contention for resources

A region with depleted NonProd headroom signals:

- Capacity strain
- Higher risk of quota exhaustion
- Less flexibility for future deployments

Placing Prod in a region with good NonProd headroom ensures the Prod workload lands in a **healthy, well-provisioned region**, not just one that happens to have Prod capacity today.

4. Overflow and Failover Capacity

In some scenarios, NonProd capacity can serve as overflow for Prod:

- **Burst capacity:** During a Prod spike, NonProd CRG might temporarily absorb overflow (subject to governance)
- **Emergency failover:** If Prod capacity is exhausted, NonProd headroom provides a safety margin
- **Multi-environment flexibility:** A region with balanced Prod + NonProd capacity can handle diverse workload mixes

By favoring regions with NonProd headroom, the model ensures Prod lands where there's **operational flexibility**, not just Prod-specific capacity.

5. Cross-Signal Validation (β Component Covers Prod Directly)

The `PS_Prod` formula already has a **Prod-specific signal**:

```

 $\beta$  component = 0.20 × Clamp(prod_crg.quota_headroom / prod_crg.quota_limit)
                        ↑
                Prod QUOTA headroom (direct Prod signal)

```

Division of labor:

- **α (NonProd headroom):** Forward-looking regional health + overflow capacity
- **β (Prod quota headroom):** Direct Prod capacity readiness

If α also used Prod headroom:

- α and β would be highly correlated (both measuring Prod capacity)
- Combined weight = $0.30 + 0.20 = 0.50$ (half the total score) dominated by Prod-only signals

- Other dimensions (distribution fairness, DR coverage, zone diversity) would be under-weighted

With α on NonProd headroom:

- α and β measure **different dimensions** of capacity health
- The formula is balanced across multiple signals, not dominated by a single capacity type

Comparison: What Changes If α Used Prod Headroom?

Current Formula (Design-of-Record)

Component	Signal	Weight	What It Measures
α	<code>non-prod_crg.effective_free / prod_crg.quantity</code>	0.30	Regional NonProd health + overflow capacity
β	<code>prod_crg.quota_headroom / prod_crg.quota_limit</code>	0.20	Direct Prod quota readiness
γ	<code>1 - prod_customer_count / total_customers</code>	0.25	Distribution fairness ⚠
δ	<code>dr_crg.coverage_ratio</code>	0.15	DR readiness
ϵ	<code>az_count / 3</code>	0.10	Zone diversity

⚠ **SPEC GAP (GAP- γ):** The `total_customers` term in the γ component is undefined in Requirements Baseline v2.5. Its source, scope, and data type are not specified. See `ac-rme_calculation_logic_comprehensive_walkthrough.md` OPEN ISSUE section for details and proposed resolution options.

Result: Multi-dimensional scoring across 5 independent signals.

Hypothetical Alternative (α Uses Prod Headroom)

Component	Signal	Weight	What It Measures
α	<code>prod_crg.effective_free / prod_crg.quantity</code>	0.30	Direct Prod capacity headroom
β	<code>prod_crg.quota_headroom / prod_crg.quota_limit</code>	0.20	Direct Prod quota headroom
γ	<code>1 - prod_customer_count / total_customers</code>	0.25	Distribution fairness ⚠️
δ	<code>dr_crg.coverage_ratio</code>	0.15	DR readiness
ϵ	<code>az_count / 3</code>	0.10	Zone diversity

⚠️ **SPEC GAP (GAP- γ):** The `total_customers` term is undefined (see note above).

Problems:

- α and β are **highly correlated** — both measure Prod capacity (combined 0.50 weight)
- **No NonProd signal** — regional capacity health is one-dimensional
- **Risk of imbalance** — Prod could land in a region with great Prod capacity but terrible NonProd capacity

Worked Example — Why It Matters

Scenario: US 3-Region Model

Request: Deploy Prod workload requiring 10,000 vCPU (64-core line item)

Region	Prod Free	Non-Prod Free	Prod Quota Headroom	α (Current)	α (If Prod)	β	PS_Prod Impact
West US 3	8,000	12,000	0.85	Clamp(12000/10000)= 1.0	Clamp(8000/10000)= 1.0	0.85	Both formulas rank this high
Central US	9,000	500	0.90	Clamp(500/10000)= 0.05	Clamp(9000/10000)= 1.0	0.90	DIVERGENCE
Canada Central	3,000	8,000	0.60	Clamp(8000/10000)= 0.80	Clamp(3000/10000)= 0.30	0.60	DIVERGENCE

Current Formula (α on NonProd):

1. **West US 3** scores highest ($\alpha=1.0$, $\beta=0.85$)
2. **Canada Central** scores 2nd ($\alpha=0.80$, $\beta=0.60$)
3. **Central US** scores **LOW** ($\alpha=0.05$) — despite high Prod capacity, its depleted NonProd headroom signals capacity strain

If α Used Prod:

1. **West US 3** still scores highest
2. **Central US** jumps to 2nd ($\alpha=1.0$, $\beta=0.90$) — looks great on Prod-only metrics
3. **Canada Central** drops ($\alpha=0.30$) — penalized for lower Prod capacity

Outcome:

- **Current formula:** Prod lands in a **balanced, healthy region** (West US 3 or Canada Central)
- **Alternative formula:** Prod might land in **Central US** — a region with Prod capacity but **starved NonProd capacity**, risking future contention

Summary — Design Rationale Table

Reason	Benefit
1. Avoid self-reference	Independent signal, not circular dependency
2. Future co-location readiness	Anticipates CVAL placement needs (2-region model)
3. Regional health indicator	Favors holistic capacity health, not just Prod-specific
4. Overflow capacity	NonProd can serve as safety margin for Prod spikes
5. Multi-dimensional scoring	α and β measure different dimensions (not correlated)

Bottom Line

Using **NonProd headroom in α** makes `PS_Prod` a **forward-looking, multi-dimensional health check**, not just a “does this region have Prod capacity right now?” binary gate.

It ensures Prod workloads land in regions with **overall capacity resilience**, not just Prod-specific headroom — anticipating future growth, overflow needs, and regional balance across all environments.

Design rationale for `PS_Prod  $\alpha$` component — ACRME Capacity Reservation baseline v2.4 · documented 28 Sep 2026